

SESSION 10 • LECTURE 8

Steady Flow Processes — Applications of the SFEE

Detailed lecture notes — device-by-device reductions with physics, five solved numericals

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Learning objectives. After this session you should be able to: (i) reduce the SFEE to the working equation of each common device by cancelling terms; (ii) compute nozzle exit velocity, turbine/compressor work and heat-exchanger duty; (iii) explain the isenthalpic nature of throttling and its use in refrigeration.

1. One Equation, Many Devices

Do not memorise a formula per device. Start every problem from the full SFEE and strike out the terms that the device physics makes negligible; what remains is the working equation.

Device	Valid assumptions	Simplified SFEE (per kg)
Nozzle / Diffuser	$q \approx 0, w = 0, \Delta PE = 0$	$(V_2^2 - V_1^2)/2 = h_1 - h_2$
Turbine	$q \approx 0, \Delta KE \approx 0, \Delta PE \approx 0$	$w = h_1 - h_2 (> 0, \text{output})$
Compressor / Pump	$q \approx 0, \Delta KE \approx 0, \Delta PE \approx 0$	$w = h_1 - h_2 (< 0, \text{input})$
Boiler / Evaporator	$w = 0, \Delta KE \approx 0, \Delta PE \approx 0$	$q = h_2 - h_1 (> 0)$
Condenser / Cooler	$w = 0, \Delta KE \approx 0, \Delta PE \approx 0$	$q = h_2 - h_1 (< 0)$
Throttling valve	$q \approx 0, w = 0, \Delta KE \approx 0$	$h_1 = h_2 \text{ (isenthalpic)}$

Device physics behind the cancellations

- Nozzle:** a shaped duct that accelerates flow; converging for subsonic flow. Pressure and enthalpy fall, velocity rises. Exit velocity: $V_2 = \sqrt{2000(h_1 - h_2) + V_1^2}$ with h in kJ/kg.
- Diffuser:** the exact reverse — decelerates flow to recover pressure (jet-engine intakes).
- Turbine:** the fluid expands against moving blades and delivers shaft work; casing is nearly adiabatic; port velocities similar.
- Compressor/pump:** shaft work is done on the fluid to raise its pressure; w emerges negative from the algebra.
- Boiler/condenser:** long tube banks with no shaft: pure enthalpy change balanced by heat.
- Throttle:** flow through a restriction (valve, porous plug, capillary): no work, negligible heat and KE change $\Rightarrow h_1 = h_2$. Pressure falls at constant enthalpy; for real gases temperature usually falls too (Joule-Thomson effect, Unit IV) — the principle of the refrigerator expansion valve. Throttling is isenthalpic but not isothermal, and highly irreversible.

KEY POINT One equation + cancellation discipline answers every steady-flow device question. State the assumptions you invoke — they carry marks.

2. Worked Examples

WORKED EXAMPLE 8.1 — Steam turbine power

Steam enters a turbine at $h_1 = 3230$ kJ/kg, $V_1 = 50$ m/s and leaves at $h_2 = 2675$ kJ/kg, $V_2 = 120$ m/s; heat loss 8 kJ/kg; ΔPE negligible; $\dot{m} = 2.5$ kg/s. Find the power developed.

Solution.

- $\Delta KE = (120^2 - 50^2)/2000 = 5.95$ kJ/kg.
- $q - w = \Delta h + \Delta KE$: $(-8) - w = (2675 - 3230) + 5.95 = -549.05$.
- $w = 541.05$ kJ/kg.
- Power = $\dot{m}w = 2.5 \times 541.05 = 1352.6$ kW ≈ 1.35 MW.

ANSWER $w \approx 541$ kJ/kg | Power ≈ 1.35 MW

Note: ΔKE (≈ 6) is tiny beside Δh (≈ 555): this justifies neglecting ΔKE in most turbine problems.

WORKED EXAMPLE 8.2 — Cooled air compressor with velocity change

Air flows at 0.5 kg/s through a compressor; h rises from 300 to 460 kJ/kg; velocity rises from 60 to 120 m/s; heat rejected 25 kJ/kg. Find the power input.

Solution.

1. $\Delta KE = (120^2 - 60^2)/2000 = 5.4$ kJ/kg.
2. $w = q - \Delta h - \Delta KE = (-25) - 160 - 5.4 = -190.4$ kJ/kg.
3. Power input $= \dot{m}|w| = 0.5 \times 190.4 = 95.2$ kW.

ANSWER $w = -190.4$ kJ/kg | Input ≈ 95.2 kW

WORKED EXAMPLE 8.3 — Nozzle exit velocity

Gas enters a nozzle at $h_1 = 3025$ kJ/kg with $V_1 = 60$ m/s and leaves at $h_2 = 2790$ kJ/kg. The flow is adiabatic. Find V_2 .

Solution.

1. $V_2 = \sqrt{(2000(h_1 - h_2) + V_1^2)}$.
2. $2000 \times 235 = 470,000$; $V_1^2 = 3600$.
3. $V_2 = \sqrt{473,600} = 688.2$ m/s.

ANSWER $V_2 \approx 688$ m/s

Note: Forgetting the 2000 gives 21.8 m/s — absurd for a nozzle; always sanity-check magnitudes.

WORKED EXAMPLE 8.4 — Condenser cooling-water requirement

Steam condenses at 1.5 kg/s from $h_1 = 2400$ kJ/kg to $h_2 = 320$ kJ/kg. Cooling water ($c_p = 4.187$ kJ/kg·K) may rise by 8 °C. Find the heat rejected and the cooling-water flow rate.

Solution.

1. Heat rejected by steam: $\dot{Q} = \dot{m}(h_2 - h_1) = 1.5 \times (320 - 2400) = -3120$ kW.
2. Water absorbs 3120 kW: $\dot{m}_w c_p \Delta T = 3120$.
3. $\dot{m}_w = 3120/(4.187 \times 8) = 3120/33.5 = 93.1$ kg/s.

ANSWER $\dot{Q} = 3120$ kW (rejected) | $\dot{m}_w \approx 93$ kg/s

Note: A heat exchanger couples two SFEE balances: heat lost by one stream = heat gained by the other.

WORKED EXAMPLE 8.5 — Throttling

Refrigerant is throttled from 900 kPa, $h = 260$ kJ/kg to 150 kPa. What is its enthalpy after the valve, and is the process isothermal?

Solution.

1. Throttle: $q \approx 0$, $w = 0$, $\Delta KE \approx 0 \Rightarrow h_2 = h_1 = 260$ kJ/kg.
2. Not isothermal in general: for real fluids T falls with p at constant h (Joule-Thomson cooling) — exactly what the refrigerator needs.

ANSWER $h_2 = 260$ kJ/kg (isenthalpic)

3. Practice Problems (answers given)

#	Problem	Answer
P1	Steam expands in a turbine from 3100 to 2580 kJ/kg at 4 kg/s, adiabatic, ΔKE negligible. Find the power.	2080 kW
P2	Air is compressed adiabatically from $h = 305$ to 480 kJ/kg. Find the work per kg.	-175 kJ/kg (input)
P3	Steam enters a nozzle at rest with $h_1 = 2980$ kJ/kg and leaves at $h_2 = 2755$ kJ/kg. Find V_2 .	≈ 671 m/s
P4	Water ($c_p = 4.187$) is heated from 30 °C to 80 °C in a boiler economiser at 2 kg/s. Find $\dot{Q}$.	418.7 kW

4. Summary

- Each device is the SFEE minus negligible terms; state your assumptions.
- Nozzle: $(V_2^2 - V_1^2)/2 = h_1 - h_2$. Turbine/compressor: $w = h_1 - h_2$. Boiler/condenser: $q = h_2 - h_1$. Throttle: $h_1 = h_2$.

- Heat exchangers couple two streams: energy lost = energy gained.
- Check magnitudes: nozzle velocities are hundreds of m/s; ΔKE is small beside Δh in work devices.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 5 (First Law Applied to Flow Processes). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.